

Phase 1 of Modern Control Project Modeling, Structural Analysis, and Validation of a NonIsothermal CSTR Reactor

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Abstract

This report presents Phase 1 of the Modern Control project focusing on a continuous stirred-tank reactor (CSTR) with nonisothermal operation. The work begins by extracting the nonlinear differential equations governing the system from the reference paper. The model is then linearized about a desired operating point, and stability and sensitivity analyses are performed. Using similarity transformations, canonical forms of the model are derived, and a comprehensive controllability and observability analysis is carried out through five different methods. Finally, the linear model is validated by comparing its response to that of the nonlinear model for step and sinusoidal inputs, with accuracy quantified using the RMSE metric. All computations are implemented in MATLAB.

1 Introduction

The CSTR is a classical chemical process exhibiting strongly nonlinear behavior due to the exponential Arrhenius reaction term and heat transfer. Accurate modeling of this system is the first step toward designing robust controllers. In this phase, after deriving the nonlinear model, linearization around the equilibrium point is performed, followed by structural, stability, and sensitivity analyses. Finally, validation of the linear model determines its accuracy and range of applicability.

2 Mathematical Model and Nonlinear State Equations

Based on the reference paper [1], the governing equations of the reactor in dimensional form are:

$$\frac{dC_a}{dt} = \frac{F}{V}(C_{ao} - C_a) - \alpha C_a e^{-E/RT} \quad (1)$$

$$\frac{dT}{dt} = \frac{F}{V}(T_o - T) - \frac{H}{\rho C_p} \alpha C_a e^{-E/RT} - \frac{UA}{\rho V C_p}(T - T_j) \quad (2)$$

$$\frac{dT_j}{dt} = \frac{F_j}{V_j}(T_{j0} - T_j) + \frac{UA}{\rho_j V_j C_{pj}}(T - T_j) \quad (3)$$

Assuming the jacket temperature dynamics are much faster than the reactor temperature dynamics, T_j can be obtained from the following algebraic equation:

$$T_j = \frac{\rho_j C_{pj} F_j T_{j0} + U A T}{\rho_j C_{pj} F_j + U A} \quad (4)$$

Substituting Eq. (4) into Eq. (2) and using the dimensionless variables introduced in the paper yields the dimensionless statespace model:

$$\frac{dx_2}{d\tau} = \frac{x_{60}}{x_1} (x_{20} - x_2) - c_0 x_2 e^{-1/x_3} \quad (5)$$

$$\frac{dx_3}{d\tau} = \frac{x_{60}}{x_1} (x_{30} - x_3) + c_1 x_2 e^{-1/x_3} - \frac{c_2 c_3 x_5 (x_3 - x_{40})}{x_1 (c_3 x_5 + c_4)} \quad (6)$$

The state, input, and output vectors are defined as:

$$x = \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} C_a / C_{aor} \\ R T / E \end{bmatrix}, \quad u = x_5 = \frac{F_j}{F_{js}},$$

$$y = x_3 \quad (\text{dimensionless reactor temperature})$$

The overall nonlinear model is $\dot{x} = f(x, u)$, $y = h(x, u)$.

Numerical values of the dimensionless parameters are given in Table 1.

Table 1: Computed dimensionless parameter values

Parameter	Value	Description
c_0	8.521062×10^{10}	Dimensionless reaction rate constant
c_1	1.109061×10^9	Dimensionless heat of reaction
c_2	25.152429	Dimensionless heat transfer
c_3	20.007080	Dimensionless jacket flow rate
c_4	241.078206	Dimensionless jacket thermal constant
x_{20}	2.040816	Dimensionless inlet concentration
x_{30}	0.035095	Dimensionless inlet temperature
x_{40}	0.035095	Dimensionless cooling water temperature
x_{60}	1	Normalized nominal inlet flow rate

3 Linearization and Stability Analysis

3.1 Finding the Equilibrium Point

For the nominal input $u_{\text{nom}} = 1$, the equilibrium point was obtained by numerically solving $\dot{x} = 0$:

$$x_{ss} = \begin{bmatrix} 1.941355 \\ 0.035537 \end{bmatrix} \implies T_{ss} = \frac{x_{3,ss} \cdot E}{R} \approx 298.413 \text{ K.}$$

3.2 Linear StateSpace Matrices

By evaluating the Jacobians of $f(x, u)$ at the equilibrium point, the linear system matrices are:

$$A = \begin{bmatrix} -1.0512 & -78.7580 \\ 0.0007 & -1.9024 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ -0.0007870 \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D = 0.$$

3.3 Stability and Transient Response

The system poles are real and given by $\lambda_1 = -1.1182$ and $\lambda_2 = -1.8354$, indicating asymptotic stability at the operating point. Since both poles are real and negative, the system is overdamped ($\zeta = 1$ for each mode), corresponding to natural frequencies of 1.12 and 1.84 rad/s and time constants of 0.894 s and 0.545 s, respectively. This implies a fast response with no oscillation and negligible overshoot. The polezero map and step response are shown in Fig. 1.

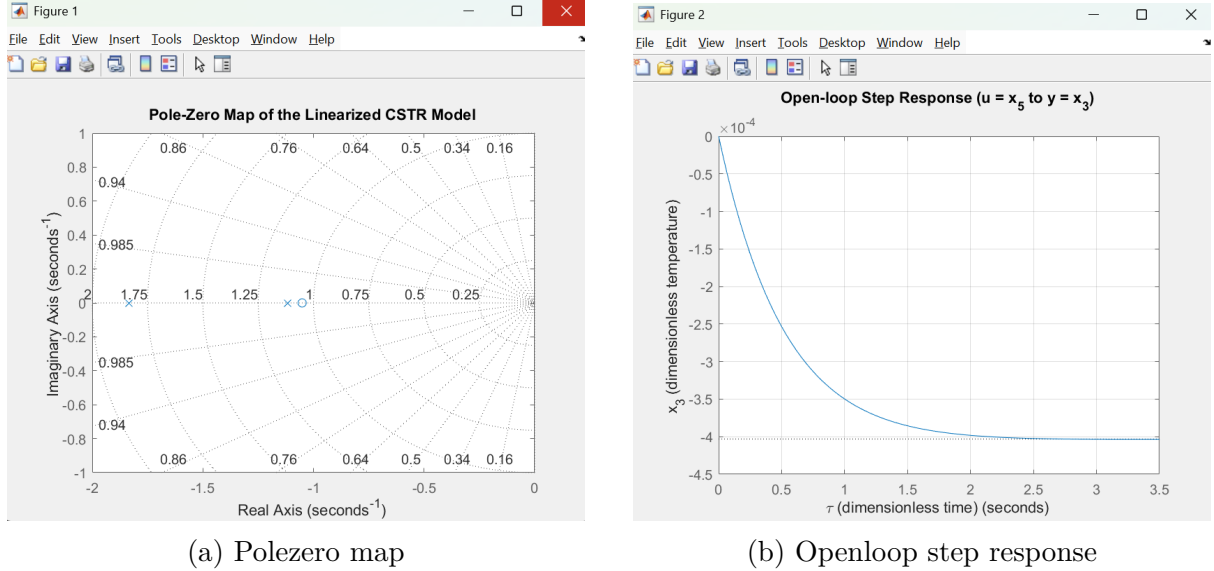


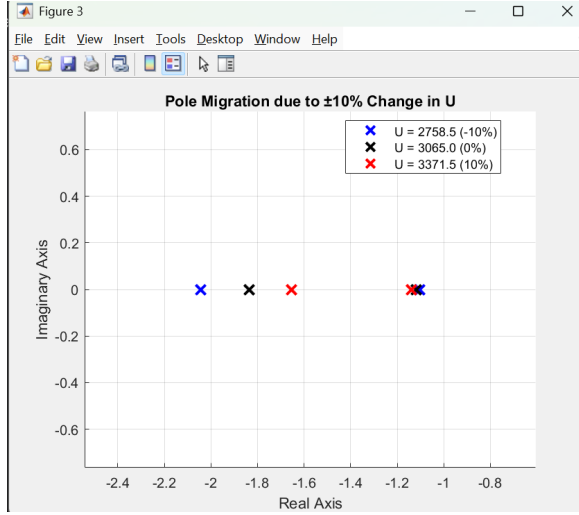
Figure 1: Stability and transient response analysis of the linearized system

3.4 Sensitivity Analysis

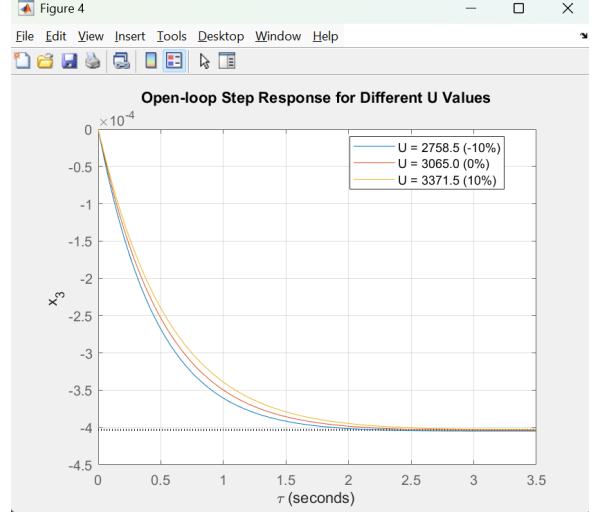
A sensitivity analysis was performed on the overall heat transfer coefficient U by varying it $\pm 10\%$. The results in Table 2 and Fig. 2 show that decreasing U (reduced cooling capacity) moves the dominant pole closer to the imaginary axis (slower response), while increasing U pushes it further into the left halfplane (faster response). Both poles remain real over the tested range, so no oscillatory behavior is introduced. This sensitivity must be considered in controller design.

Table 2: Dominant pole location with variations in U

U (kJ/h m ² K)	Pole	Comment
2758.5 (−10%)	−1.1047	Slower response
3065.0 (nominal)	−1.1182	Nominal
3371.5 (+10%)	−1.1377	Faster response



(a) Pole migration



(b) Step response comparison

Figure 2: Sensitivity analysis results for parameter U

4 Canonical Forms and Structural Analysis

4.1 Canonical Forms and Similarity Transformations

After obtaining the system matrices (A, B, C, D) , similarity transformations were used to convert the model to the controller and observer canonical forms. The corresponding transfer function is:

$$G(s) = \frac{-0.000787017 s - 0.000827338}{s^2 + 2.9536 s + 2.05235}.$$

The controller canonical form is:

$$A_{Co} = \begin{bmatrix} 0 & 1 \\ -2.0523 & -2.9536 \end{bmatrix}, \quad B_{Co} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

$$C_{Co} = \begin{bmatrix} -0.0008273 & -0.0007870 \end{bmatrix}, \quad D_{Co} = 0.$$

The observer canonical form is:

$$A_{Ob} = \begin{bmatrix} 0 & -2.0523 \\ 1 & -2.9536 \end{bmatrix}, \quad B_{Ob} = \begin{bmatrix} -0.0008273 \\ -0.0007870 \end{bmatrix},$$

$$C_{Ob} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \quad D_{Ob} = 0.$$

These forms clarify the internal structure of the system for further analysis.

4.2 Comprehensive Controllability Analysis

Controllability was verified using five independent methods, all in complete agreement:

1. **Controllability matrix:** $\mathcal{C} = [B, AB]$ was formed with entries $\begin{bmatrix} 0 & 0.0620 \\ -0.0008 & 0.0015 \end{bmatrix}$ and rank 2.
2. **Manual calculation:** The same matrix was computed without builtin MATLAB functions, and its rank was confirmed to be 2.
3. **Jordan form:** In the Jordan form $J = \text{diag}(-1.8354, -1.1182)$ with $B_J = \begin{bmatrix} -0.0008605 \\ 0.0000735 \end{bmatrix}$, all rows of B_J corresponding to each Jordan block are nonzero.
4. **Controllability Gramian:** $W_c = \begin{bmatrix} 0.0003169 & -0.0000042 \\ -0.0000042 & 0.0000002 \end{bmatrix}$, with eigenvalues $\{0.0000001, 0.0003170\}$, confirming W_c is positive definite.
5. **PopovBelevitchHautus test:** For both eigenvalues $\lambda_1 = -1.1182$ and $\lambda_2 = -1.8354$, the matrix $[A - \lambda_i I \ B]$ has full rank (2).

4.3 Comprehensive Observability Analysis

Observability was also confirmed using five methods:

1. **Observability matrix:** $\mathcal{O} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0.0007 & -1.9024 \end{bmatrix}$ has rank 2.
2. **Manual calculation:** The rank of $\mathcal{O}$ was manually verified to be 2.
3. **Jordan form:** In J , the associated $C_J = \begin{bmatrix} 1 & 1 \end{bmatrix}$, so all columns corresponding to each Jordan block are nonzero.
4. **Observability Gramian:** $W_o = \begin{bmatrix} 0.0000 & 0.0001 \\ 0.0001 & 0.2604 \end{bmatrix}$, with eigenvalues $\{0.0000, 0.2604\}$, confirming W_o is positive (semi)definite.
5. **PopovBelevitchHautus test:** For both $\lambda_1 = -1.1182$ and $\lambda_2 = -1.8354$, $\begin{bmatrix} A - \lambda_i I \\ C \end{bmatrix}$ has full rank (2).

Conclusion: The system is both controllable and observable, hence it is a **minimal realization**.

5 Validation and Model Comparison

To assess the accuracy of the linear model, its response is compared with that of the original nonlinear model for step and sinusoidal inputs. The quantitative metric is the root mean square error (RMSE).

5.1 Time Response Comparison

Figure 3 shows an example of the response comparison for a sinusoidal input. The close agreement between the two curves indicates the high accuracy of the linear model in the vicinity of the operating point.

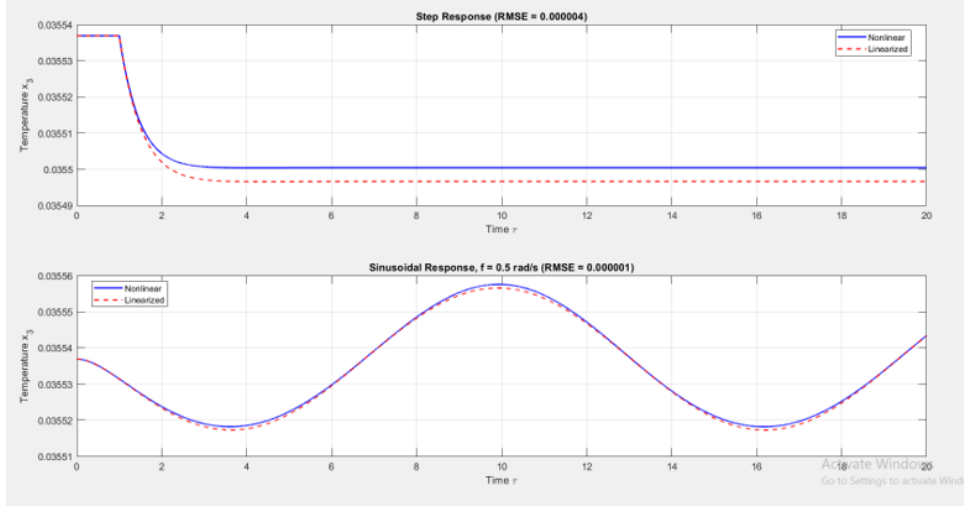


Figure 3: Comparison of linear and nonlinear model responses to a sinusoidal input with amplitude 0.1 and frequency 0.5 Hz

5.2 Error Quantification and Comprehensive Analysis

For the representative test cases, the computed errors were:

- **Step input:** $\text{RMSE} = 4 \times 10^{-6}$, indicating excellent agreement between the linear and nonlinear models.
- **Sinusoidal input:** $\text{RMSE} = 1 \times 10^{-6}$, again showing very close tracking between the two models.

Figure 4 shows how the RMSE varies with input amplitude and frequency. As amplitude increases (deviation from the operating point grows), the error generally increases for both input types; for the sinusoidal case, error tends to decrease or stay roughly constant as frequency rises, a consequence of the fast system dynamics.

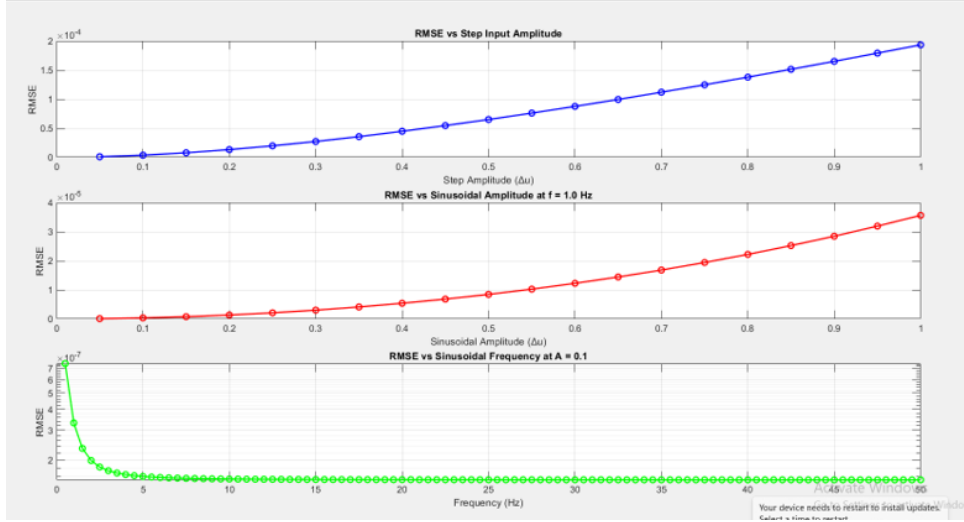


Figure 4: RMSE variation with step amplitude, sinusoidal amplitude, and frequency

5.3 Final Model Validity Assessment

The results clearly show that the obtained linearized model is an extremely accurate approximation of the nonlinear system dynamics around the operating point. The very small RMSE and the matching response curves fully validate this model for designing linear controllers (such as LQR or PID) in Phase 2 of the project. It should be noted that as the system moves further from the operating point (excessive input amplitudes), nonlinear effects intensify and the linear model's accuracy degrades.

6 Overall Conclusion

In this phase, the essential and critical steps for analyzing and preparing the model of a nonlinear system were completed. The nonlinear CSTR model was successfully derived and linearized around the operating point. Stability analysis confirmed asymptotic stability with two real, negative poles, and sensitivity analysis provided valuable insight into the influence of physical parameters on system dynamics. Comprehensive analyses proved the system to be a minimal realization. Finally, validation using the RMSE metric demonstrated that the linear model possesses very high accuracy and can serve as a reliable foundation for designing advanced control systems in Phase 2.

References

- [1] Perez, T., & Albertos, P. (2004). Self-oscillating and chaotic behaviour of a PI-controlled CSTR. *Journal of Process Control*, 14(3), 321-330.